End-To-End Memory Networks

Motivation. Many reasoning tasks require models to consult stored knowledge before producing an answer. Traditional neural networks compress all information into hidden states, which limits their ability to retrieve specific facts.

End-to-End Memory Networks introduce an explicit memory component that can be queried through attention.

Architecture. The model consists of three main components:

- An input representation
- A memory module
- An output module

The memory stores a set of vectors representing facts or observations:

$$m_i \in \mathbb{R}^d$$

A query vector u represents the current question.

Memory Attention. The model computes attention weights over memory entries:

$$p_i = \text{Softmax}(u^\top m_i)$$

These weights determine which facts are relevant.

A response vector is computed as

$$o = \sum_i p_i c_i$$

where c_i are output embeddings associated with each memory entry.

Multi-Hop Reasoning. The model can perform multiple reasoning steps ("hops").

After each hop, the query is updated:

$$u_{k+1} = u_k + o_k$$

This allows the model to iteratively refine its reasoning and combine information from multiple memory entries.

Prediction. The final prediction is produced via

$$\hat{a} = \text{Softmax}(W(u_K + o_K))$$

where K is the number of hops.

Applications. The model was evaluated on the bAbI dataset, which contains synthetic reasoning tasks such as:

- question answering
- path finding
- logical deduction

Multi-hop memory access allows the model to combine several supporting facts.

Advantages. Compared to earlier memory networks, this architecture:

- is fully differentiable
- can be trained end-to-end with gradient descent
- does not require supervision of memory retrieval steps

Limitations. Several challenges remain:

- Memory size must remain relatively small
- Attention mechanisms scale poorly with large knowledge bases
- Performance degrades on more complex reasoning tasks

These limitations motivated later architectures such as transformers and retrieval-augmented models.

Takeaway. End-to-End Memory Networks introduced differentiable attention over explicit memory, enabling neural networks to retrieve and combine facts dynamically. This work strongly influenced later attention-based architectures and can be viewed as an early precursor to modern transformer models.